

ACM RECSYS 2026 · RECSYS CHALLENGE WORKSHOP

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

Nidhin Pattaniyil · Semih Yagli · Tanwir Zaman

Minneapolis · 2 October 2026 · Music-CRS track, team npatta01

47,071

tracks in the catalog

11

ways to search

1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of numerous vertical bars of varying heights in a dark gray color.

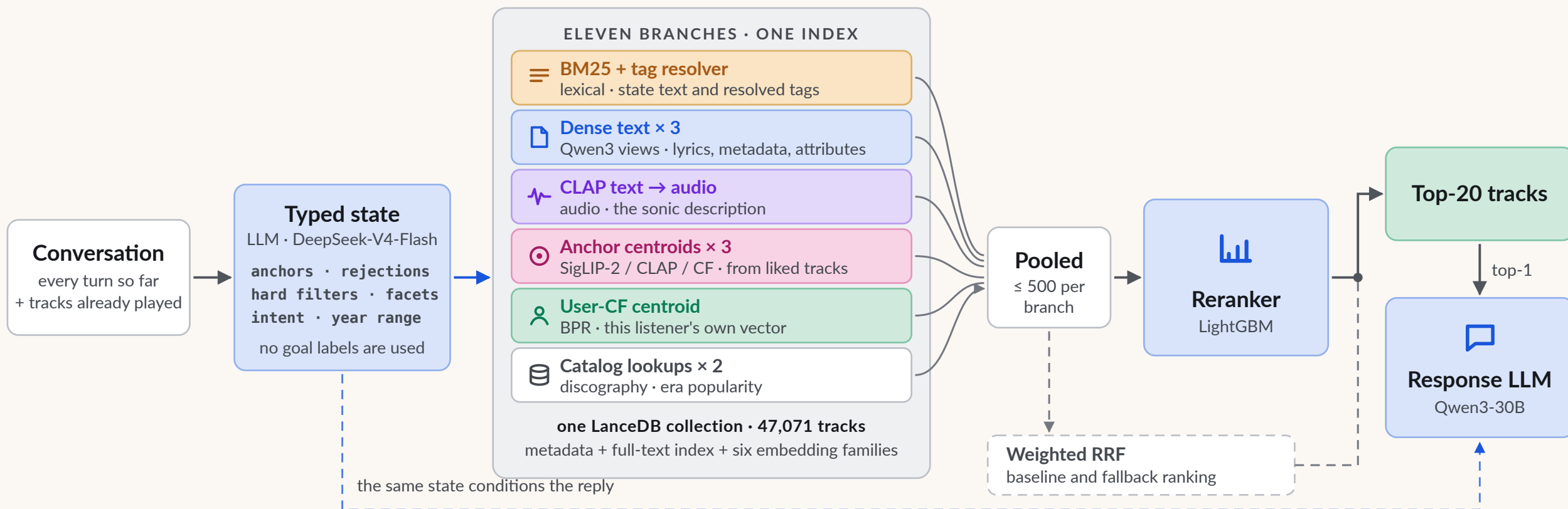


Contents

1	How it works	the state, retrieval, reranking, the reply
2	How it scored	the official results
3	Why we did not score higher	labels, training, ranking, missing data, the reply
4	Conclusion	what we showed, and questions

≡ One state, eleven branches, one learned ranker

APPROACH



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.

Request understanding

APPROACH

WE HAD JUST PLAYED



Ryan Adams & The Cardinals — “Let Us Down Easy”

LISTENER

...more tracks with that kind of emotional depth and acoustic focus, maybe branching out to other singer-songwriter or alt-country artists?

LLM



REQUEST TYPE	new artist
AVOID	Ryan Adams & The Cardinals
MOOD	introspective, emotional depth
SOUND	acoustic-driven
GENRE	singer-songwriter, alt-country

The LLM reads **every turn so far**: “other artists” after a Ryan Adams track becomes an artist to avoid, and the mood carries over.



What the state can hold

APPROACH

THE REQUEST

- request type
- intent mode
- routing flags

WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

WHAT IT IS ABOUT

- lyrical theme

BOUNDS

- release-year window
- process constraints



How the state drives retrieval

APPROACH



The request

request type · intent mode · routing flags



Which branches fire, and how much each one weighs in the merge

routing weights

pivot skips anchors

gates discography



What they like

liked tracks · named artists and tags



Search near liked tracks; match named entities

anchor centroids ×3

BM25 fields

artist discography



What it sounds like · is about

mood · genre · sonic · lyrical theme



The text, audio and lyrics queries

dense text ×3

CLAP text→audio

lyrics view



What they ruled out

rejected tracks and artists



Dropped from every branch before the ranker

hard drop

same-artist demote



Bounds

release-year window · date filters



Mask out-of-range tracks; boost the era

candidate mask

era-popularity lookup

Retrieval — eleven branches

APPROACH



×1

BM25 + tag resolver

keywords and resolved tags
from the state



×3

Dense text

Qwen3 embeddings of lyrics,
metadata, attributes



×1

CLAP text → audio

the sonic description, matched
to audio



×3

Anchor centroids

near liked tracks: cover art,
audio, listening



×1

User-CF centroid

this listener's own BPR vector



×2

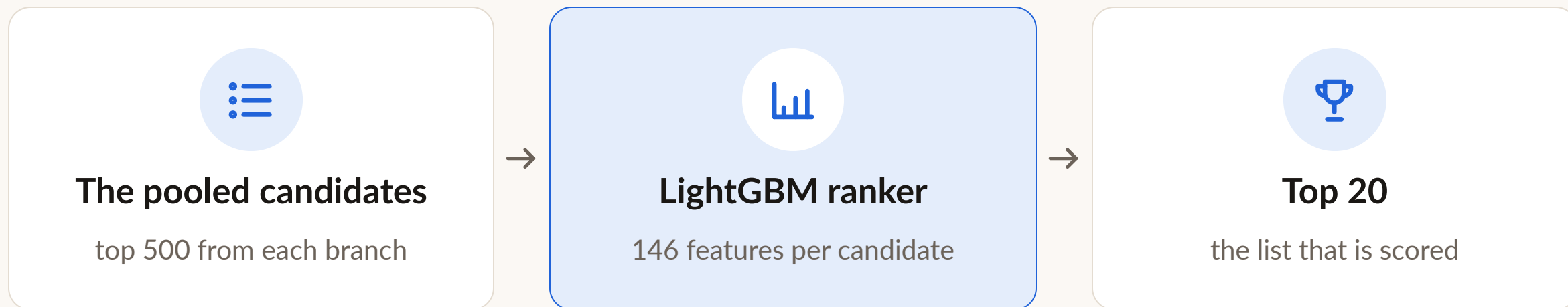
Catalog lookups

the artist's discography; era
popularity

Each branch returns its **top 500**; the ranker sees their union.

Reranking

APPROACH





What the ranker leans on

APPROACH

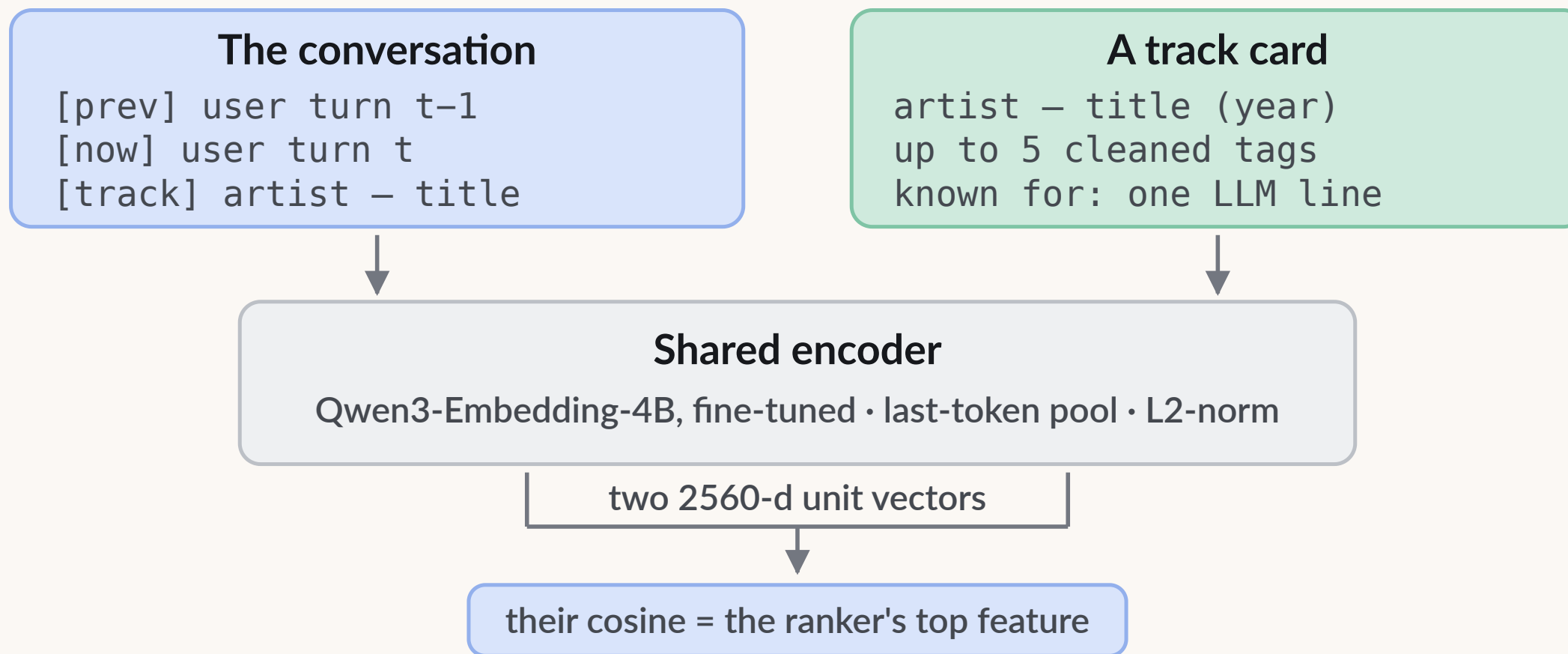


Share of the ranker's split gain, by feature family. One signal — the bi-encoder cosine — carries **59%**; constraints and rejections carry **0.7%**.



The bi-encoder behind that top feature

APPROACH





Response generation

APPROACH

INPUTS

Conversation

"...it feels a bit too serious. I'm looking for something more uplifting and positive..."

Latest state

Mood **uplifting, positive**

Avoid **serious**

Top-ranked track

Track **Röyksopp – "Happy Up Here"**

Tags **cheerful, hopeful, electronic...**



Qwen3-30B

temperature 0 • one
pass

1-2 sentences about
the top track only



OUR REPLY

"Perfect – 'Happy Up Here' by Röyksopp is exactly the kind of uplifting, cheerful electronic track you're looking for. With its bright synths and hopeful groove, it's designed to lift your spirits and make you smile."

✓ How it scored

RESULTS

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B · final	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

catalog and reply-wording
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



Why we did not score higher

WHAT IS NOT IDEAL

- | | | |
|---|---|---|
| 1 | The answer keys are inconsistent | they often keep the artist the listener just left |
| 2 | We trained and checked on the same turns | no clean held-out set |
| 3 | The ranker did not act on the state | rejections carried 0.7% of its decisions |
| 4 | No data from outside the catalog | no tempo, key or soundtrack links |
| 5 | Gaps in our response generation | the reply was never checked |



Label example — “from a different artist”

WHAT IS NOT IDEAL

RECOMMENDER AGENT PLAYED



Bonobo — “Cirrus”

LISTENER, NEXT TURN

...something with a similar chill, electronic vibe, maybe with a subtle beat, but **from a different artist?**

ANSWER KEY FOR THAT TURN



Bonobo — “Jets”

MOVES TOWARD GOAL

× Our judges: wrong — same artist

NEXT TURN

“Jets” is cool, but I was actually hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

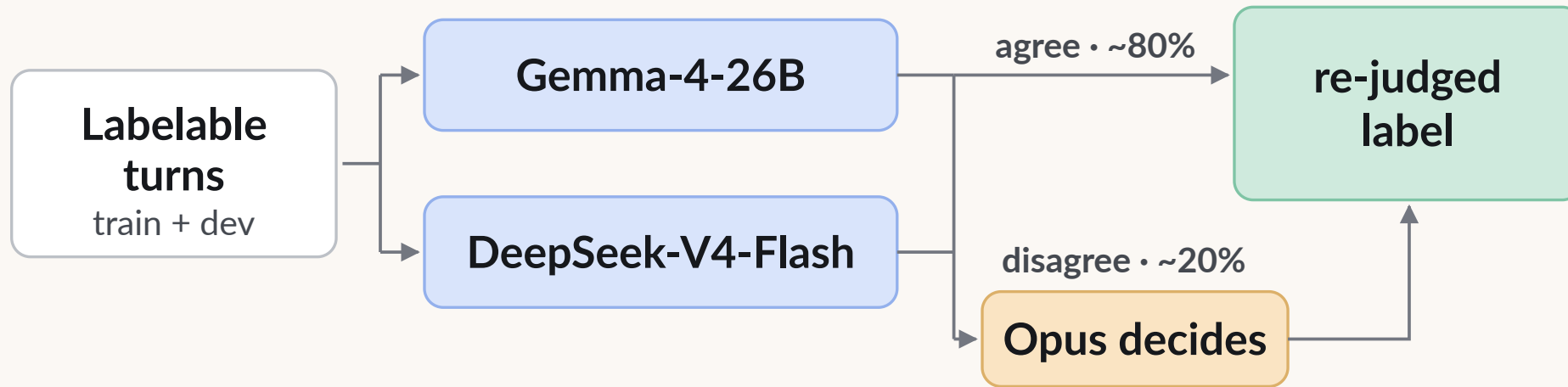
× Our judges: wrong — same artist

Twice asked to leave Bonobo; both answer keys are Bonobo, **marked helpful**.



How we relabelled, and what we found

WHAT IS NOT IDEAL



17%

of training turns keep the artist the listener had just asked to leave

18,222 turns

113,393

turns re-judged

every labelable train and dev turn — released

Retrained on these labels, the ranker scored lower **against the original labels** — a test that could never favour the fix.



Why we trained on the dev set

LESSONS



State extraction: 1 paid LLM call per turn

to read each turn into a state



Too costly to extract state for 121.6k training turns

so no ranker features there



Trained on the 8,000 dev turns

the same turns we checked it on

Dev, turns it trained on



0.384

Blind-B, official



0.254

nDCG@20. The dev row is an earlier model version, shown because it is the number we selected on. Source: paper §4.3, §5.3.



Extracted, but the ranker did not act on it

WENT WRONG



*"I gotta break away from **Deltron** and **Del** completely"*

Dead Prez — Hip Hop

state held both artists as avoid

NOT ACTED ON



*"branching out to **other** singer-songwriter artists"*

Ryan Adams — When The Stars Go Blue

state held Ryan Adams as avoid

NOT ACTED ON



*"still focusing on **Juanes**... haven't found the depth I'm after"*

Juanes — Me Enamoré De Ti

state held Juanes as rejected

NOT ACTED ON

We wanted the ranker to use what the state recorded; it served that artist anyway. We think the labels taught it to: they often repeat the just-played artist, so rejections carried only **0.7%** of its decisions.

No data from outside the catalog

WHAT IS NOT IDEAL

A tempo and a key

“exactly 126.70 bpm, in G minor”



the catalog has no tempo or key field
no branch can match on either

A soundtrack

*“a specific 80s synth-pop song from the **Breaking Bad** soundtrack”*



we served Soft Cell — “Tainted Love”
no film or TV links in the catalog

The state was right both times. The gap is **knowledge, not understanding**.



Gaps in our response generation

WHAT IS NOT IDEAL

The track is not in the catalog, so retrieval could not return it — the reply presents our substitute as if it were the request

1

"I wanted 'Watercolors' by Pat Metheny specifically."

"Got it — here's 'Alfie' by Pat Metheny..."

NOT FLAGGED

2

"Could you please play Sarah McLachlan's 'Fallen'?"

"You're in the right place — 'Hold On' by Sarah McLachlan is a perfect match..."

NOT FLAGGED

Our pick broke the listener's constraint — the reply recommends it anyway

3

*"...something from a **different artist** this time"*

"...while this track is by him, it's from a different era and band..." — Ryan Adams again

NOT FLAGGED

4

*"...we're still focusing on **Juanes**..."*

"I know you're looking to explore beyond Juanes, but 'Me Enamoré De Ti' is a standout..."

NOT FLAGGED

One pass, no check: the reply could flag **neither unavailable tracks nor unmet constraints**. The prompt was chosen on Blind-A (judge 4.70 there, 3.30 on Blind-B).

Conclusion

WHAT WE SHOWED



A typed conversation state

One LLM call per turn records the request, likes, rejections, sound and bounds.



Retrieval → ranking → reply

The state drives eleven retrieval branches, a learned ranker, and a reply about the top track.



Gaps in our system

No knowledge from outside the catalog; trained and checked on the same turns; the reply was never checked.



Inconsistent labels

17% of training turns keep a rejected artist — we release 113k re-judged labels.

Thank you – questions?



Paper, poster, slides, code, models, data

npatta01.github.io/music-crs-2026